Week 8: Relationship Between Library and Information Science (LIS) and Other Fields of Study

Introduction

Library and Information Science (LIS) is inherently interdisciplinary. It borrows theories, methods, and technologies from multiple fields—and, in turn, contributes standards, services, and ethical frameworks that benefit those fields. Understanding these relationships helps students see where LIS professionals add value outside the traditional library and how collaboration improves teaching, research, innovation, and community development.

Why LIS Is Interdisciplinary

LIS manages the full information lifecycle—creation, description, storage, discovery, access, use, and preservation. Each stage intersects with other disciplines: computer science provides systems and algorithms; law frames copyright and privacy; education shapes information literacy pedagogy; management steers service delivery; and the social sciences explain human information behavior. This makes LIS a natural bridge across academia, government, industry, and communities.

Key Relationships

a) LIS and Computer Science / Information Technology

How they connect: Library systems (OPACs/ILS/LMS), digital repositories, discovery layers, metadata harvesters (OAI-PMH), APIs, search/recommendation engines, authentication (SSO), and cybersecurity are built with CS/IT tools.

LIS → CS/IT: Domain expertise in metadata (MARC, Dublin Core), ontologies, authority control, user needs, and evaluation of search relevance.

CS/IT → LIS: Databases, IR algorithms, NLP, UX engineering, cloud platforms, and automation.

Example: Migrating a university library to a cloud ILS and integrating it with the learning management system so students authenticate once and access e-resources seamlessly.

b) LIS and Data Science / Research Data Management (RDM)

How they connect: LIS curates research data, writes Data Management Plans (DMPs), ensures FAIR principles (Findable, Accessible, Interoperable, Reusable), and supports reproducible science.

LIS → DS: Preservation strategies, metadata for datasets (e.g., DataCite), persistent identifiers (DOIs, ORCID), and documentation best practices.

DS → LIS: Analytics for collection development, usage dashboards, and evidence-based decision-making.

Example (Nigeria): A librarian partners with a public health lab to curate malaria datasets in an institutional repository and train researchers on data citation.

c) LIS and Archives / Records Management

How they connect: Overlapping missions—evidence, memory, accountability. Records managers ensure reliable records during active use; archivists preserve permanently valuable records. Librarians contribute discovery and access services.

LIS → Archives/RM: User-focused discovery, subject access, and community outreach.

Archives/RM → LIS: Appraisal, provenance, authenticity, and legal admissibility of records.

Example: Implementing an enterprise records schedule in a ministry, then transferring permanent records to a digital archive with public access policies.

d) LIS and Law (Copyright, Licensing, Privacy, FOI)

LIS 111 NOT FOR SALE

How they connect: Libraries navigate copyright exceptions (fair dealing), licenses for e-resources, data protection, and freedom of information.
LIS → Law: User education on ethical use, copyright clearance workflows, and open licensing (Creative Commons).
Law → LIS: Legal frameworks that shape access, digitisation, and privacy policies.
Example: Negotiating a consortial e-journal license for multiple universities while ensuring remote access and text-mining rights.

e) LIS and Education / Instructional Design
How they connect: Information literacy is core to lifelong learning. Librarians co-design curricula, embed IL modules, and assess learning outcomes.
LIS → Education: Pedagogies for evaluating sources, avoiding plagiarism, and using databases.
Education → LIS: Learning theories, assessment strategies, and inclusive teaching practices.
Example: An embedded librarian teaches a graded module on systematic searching in a postgraduate research methods course.

f) LIS and Communication / Media & Journalism
How they connect: Both care about credible information, audience needs, and dissemination.
LIS → Media: Verification, databases for investigative reporting, fact-checking workflows, media archiving.
Media → LIS: Storytelling techniques and outreach that boost library impact.
Example: Collaborating with campus radio to run weekly segments on misinformation and digital safety.

g) LIS and Management / Public Administration
How they connect: Libraries are service organisations needing strategic planning, budgeting, HR, marketing, and quality assurance.
LIS → Management: Evidence-based service design using usage data and community needs assessments.
Management → LIS: Project management (Agile), performance indicators, and policy development.
Example: A library develops a 3-year strategic plan tied to university KPIs and SDG 4 (Quality Education).

h) LIS and the Social Sciences (Sociology, Psychology, Anthropology)
How they connect: Human information behavior (HIB), digital divides, community informatics, and cultural practices.
LIS → Social Sciences: Instruments for studying search behavior and information needs.
Social Sciences → LIS: Theories explaining motivation, trust, cognition, and social networks.
Example: A study of farmers' information practices informs an agricultural extension library's mobile outreach.

i) LIS and Health Sciences (Health Informatics)
How they connect: Clinical librarianship, systematic reviews, evidence synthesis, patient education.
LIS → Health: Search strategies (MeSH), appraisal tools, knowledge translation.
Health → LIS: Standards (HL7/FHIR), privacy, and clinical workflows.
Example: A medical librarian co-authors a Cochrane-style review and trains residents on PubMed Clinical Queries.

LIS 111 NOT FOR SALE

j) LIS and the Humanities / Digital Humanities
How they connect: Digitisation, TEI encoding, text mining, and cultural heritage discovery.
LIS → DH: Metadata schemas, preservation, and discovery platforms.
DH → LIS: Methods for computational analysis and community-engaged curation.
Example: Digitising oral histories in local languages and building a searchable website with annotations.

k) LIS and Geography / GIS
How they connect: Spatial discovery layers for collections; mapping community needs and service coverage.
Example: Mapping school library access across a state to guide resource allocation.

l) LIS and Business / Entrepreneurship
How they connect: Market intelligence, competitive analysis, knowledge management (KM), and patent searching.
Example: A public library's "business information clinic" helps SMEs find standards, tenders, and market data.

m) LIS and Philosophy / Ethics
How they connect: Intellectual freedom, privacy, equity of access, algorithmic transparency, and cultural sensitivity.
Example: Drafting an AI use policy for library services grounded in fairness and accountability.

n) LIS and Linguistics / Terminology
How they connect: Controlled vocabularies, ontologies, and multilingual access.
Example: Building a bilingual thesaurus to improve discovery of indigenous knowledge resources.

Summary
LIS is a connective discipline that amplifies the work of many fields by organising knowledge, enabling discovery, safeguarding rights, and teaching people how to use information wisely. Collaboration with CS/IT, data science, health, education, law, management, media, and the social sciences makes LIS more powerful—and makes those fields more humane, ethical, and efficient.

LIS 111 NOT FOR SALE